Use Case Descriptions for: *(1) Connect 4, (2) Tic Tac Toe, (3) Checkers*

Use Case: Connect 4 Home Menu

Iteration: 1

Primary Actor: Player

Goal In Context: Allow the player to access the main Connect 4 game interface and select options such as starting a new game, viewing the leaderboard, or joining a queue for matchmaking.

Preconditions:

- The player is logged in and has access to their profile.
- The Connect 4 game module is available.

Trigger: The player navigates to the Connect 4 game section from the Player Profile Home Menu.

Scenario:

1. The player selects “Connect 4” from the Player Profile Home Menu.
2. The system loads the Connect 4 Home Menu.
3. The player sees options to view the Connect 4 leaderboard, or join the matchmaking queue.
4. The player selects their desired option (e.g., “Join Queue”).
5. The system directs the player to the chosen option.

Post Conditions: The player has accessed their chosen option within the Connect 4 module.

Exceptions:

- **E1:** The Connect 4 module is unavailable. The system shows an error message and redirects the player to the Player Profile Home Menu.

Priority: High

When Available: Upon player login and successful navigation to Connect 4.

Frequency of Use: Frequent, as this is the entry point for Connect 4-related activities.

Channel to Actor: User Interface (UI)

Secondary Actors: None

Channels to Secondary Actors: N/A

Open Issues:

- Should the Connect 4 Home Menu include additional options, such as game settings or tutorials? Or play with AI? Or local multiplayer?

Use Case: Connect 4 Game Queue

Iteration: 1

Primary Actor: Player

Goal In Context: To allow the player to enter a queue for matchmaking in order to find an opponent for a Connect 4 game.

Preconditions:

- The player has accessed the Connect 4 Home Menu.
- The matchmaking service is active.

Trigger: The player selects the option to join the Connect 4 game queue.

Scenario:

1. The player selects “Join Queue” from the Connect 4 Home Menu.
2. The system adds the player to the Connect 4 game queue.
3. The system continuously searches for a compatible opponent.
4. If an opponent is found, the system notifies the player and initiates the Connect 4 Game Instance.

Post Conditions: The player has either joined a match or is informed that no match is currently available.

Exceptions:

- **E1:** No players are available in the queue. The system displays a “No Match Found” message.
- **E2:** Matchmaking service is down. The system shows an error message and returns the player to the Connect 4 Home Menu.

Priority: High

When Available: Anytime the player chooses to enter the queue.

Frequency of Use: Moderate, dependent on player activity and matchmaking wait times.

Channel to Actor: UI notification (possibly with a “Searching for opponent…” indicator)

Secondary Actors: Matchmaking service

Channels to Secondary Actors: System-to-system communication with matchmaking service

Open Issues:

- What criteria are used to find a compatible opponent (e.g., skill level)?
- Should there be an estimated wait time shown to the player?

Use Case: Connect 4 Game Instance

Iteration: 1

Primary Actor: Player

Goal In Context: Allow the player to participate in a Connect 4 game against an opponent.

Preconditions: A match has been successfully created, and an opponent has been assigned.

Trigger: The system initiates the game instance upon finding an opponent for the player in the queue.

Scenario:

1. The system matches the player with an opponent and starts the game instance.
2. The player is notified that the game is starting and is presented with the Connect 4 game board.
3. The player and the opponent take turns making moves until the game ends (either by winning, losing, or a draw).
4. The system records the game result and updates the leaderboard.
5. The player is shown the game result screen (victory, defeat, or draw).

6. The player has the option to return to the Connect 4 Home Menu or view the leaderboard.

Post Conditions: The game results are recorded, and the player is directed to the game aftermath options.

Exceptions:

- **E1:** Opponent disconnects. The player is informed, and the game ends prematurely.
- **E2:** Connection loss. The system saves the game state temporarily or informs the player of the disconnection.

Priority: High

When Available: Once matchmaking is complete.

Frequency of Use: Dependent on player's success in finding a match.

Channel to Actor: UI with real-time updates

Secondary Actors: Opponent player

Channels to Secondary Actors: Network communication for real-time game moves

Open Issues:

- Should there be an option to rematch with the same opponent?
- How is the game state handled if a player disconnects?

Use Case: No Match Found

Iteration: 1

Primary Actor: Player

Goal In Context: Notify the player that no match is currently available in the queue and provide options to retry or exit.

Preconditions: The player has joined the Connect 4 Game Queue.

Trigger: The system fails to find an opponent within a defined timeout period or due to low player availability.

Scenario:

1. The player joins the Connect 4 Game Queue.
2. The system continuously searches for an opponent.
3. After a specified timeout period, the system fails to find a match.
4. The system displays a “No Match Found” notification to the player.
5. The player is given options to retry joining the queue or to return to the Connect 4 Home Menu.

Post Conditions: The player is informed that no match was found and can choose to retry or exit.

Exceptions:

- **E1:** System experiences an error in retrieving player queue status. An error message is shown, and the player is redirected to the Connect 4 Home Menu.

Priority: Medium

When Available: During periods of low player activity or when the matchmaking queue has a timeout.

Frequency of Use: Occasional, depending on player availability and queue demand.

Channel to Actor: UI notification

Secondary Actors: None

Channels to Secondary Actors: N/A

Open Issues:

- How long should the system wait before displaying the “No Match Found” message?
- Should the player receive an estimated wait time or indication of current queue length?

Use Case: Connect Four Leaderboard

Iteration: 1

Primary Actor: Player

Goal In Context: To allow the player to view the leaderboard for Connect Four, displaying essential rankings and game-related statistics of top players.

Preconditions:

- The player has accessed the Connect Four Home Menu.
- Leaderboard data is available and up-to-date.

Trigger: The player selects the “Leaderboard” option from the Connect Four Home Menu.

Scenario:

1. The player navigates to the Connect Four Home Menu.
2. The player selects “Leaderboard.”
3. The system retrieves the latest leaderboard data for Connect Four.
4. The system displays the leaderboard, showing the following statistics for each player:
    - **Rank**: The player’s position based on win rate or score.
    - **Username**: The display name of each ranked player.
    - **Total Wins**: The total number of games the player has won.
    - **Win Rate**: The percentage of games won out of total games played.
    - **Longest Win Streak**: The highest number of consecutive wins achieved.
    - **Games Played**: The total number of games the player has participated in.
    - **Last Played Date**: The date of the player’s most recent game.
    - **Moves to Win (Average)**: The average number of moves the player needed to win a game.
5. The player can scroll through or search the leaderboard to find specific players or rankings.
6. The player can filter the leaderboard based on categories like “Top Wins” or “Highest Win Rate.”
7. The player can return to the Connect Four Home Menu or view their own profile from the leaderboard page.

Post Conditions: The player has successfully viewed the Connect Four leaderboard and can navigate back to other Connect Four options.

Exceptions:

- **E1:** The leaderboard data fails to load due to a network issue. The system displays an error message and provides an option to retry or return to the Connect Four Home Menu.

- **E2:** The leaderboard service is temporarily unavailable. The system informs the player and directs them to try again later or return to the Connect Four Home Menu.

Priority: Medium

When Available: Anytime the player is in the Connect Four Home Menu.

Frequency of Use: Occasional, based on the player's interest in checking rankings and scores.

Channel to Actor: UI display on the leaderboard screen

Secondary Actors: Leaderboard database/service

Channels to Secondary Actors: System-to-database communication to retrieve leaderboard information

Open Issues:

- Should the leaderboard allow additional filtering options (e.g., friends, region)?
- How frequently should the leaderboard data be refreshed for accuracy?

Use Case: Tic Tac Toe Home Menu

Iteration: 1

Primary Actor: Player

Goal In Context: To enable the player to access the main game interface for Tic Tac Toe and select options such as starting a new game, viewing the leaderboard, or joining the matchmaking queue.

Preconditions:

- The player is logged in and has access to their profile.
- The Tic Tac Toe module is active.

Trigger: The player navigates to the Tic Tac Toe game section from the Player Profile Home Menu.

Scenario:

1. The player clicks “Tic Tac Toe” from the Player Profile Home Menu.
2. The system loads Tic Tac Toe Home Menu.
3. Player is given options to either view the Tic Tac Toe leaderboard or join the matchmaking queue.
4. The player selects their desired option (e.g., “Join Queue”).
5. The system directs the player to the chosen option.

Post Conditions: The player is successfully able to access their chosen feature in the Tic Tac module.

Exceptions:

- **E1:** The Tic Tac Toe module is unavailable. The system shows an error message and redirects the player to the Player Profile Home Menu.

Priority: High

When Available: When player is logged in & navigates to Tic Tac Toe

Frequency of Use: Frequent, as this is the starting point for Tic Tac Toe-related activities.

Channel to Actor: User Interface (UI)

Secondary Actors: None

Channels to Secondary Actors: N/A

Open Issues: N/A

Use Case: Tic Tac Toe Game Queue

Iteration: 1

Primary Actor: Player

Goal In Context: To allow the player to enter a matchmaking queue to find an opponent for the Tic Tac Toe game.

Preconditions:

- The player has accessed the Tic Tac Toe Home Menu.
- Matching making feature is working.

Trigger: The player selects the option to join the Tic Tac Toe game queue.

Scenario:

1. The player selects “Join Queue” from the Tic Tac Toe Home Menu.
2. The system adds the player to the Tic Tac Toe game queue.
3. The system searches for a opponent.
4. Once opponent is found, system notifies the player and initiates the Tic Tac Toe Game Instance.

Post Conditions: Player either joins a match or gets a message that a match is not currently available.

Exceptions:

- **E1:** No players are available in the queue, so the system displays a “No Match Found” message.
- **E2:** Matchmaking service is offline or is malfunctioning so the system shows an error message and returns the player to the Tic Tac Toe Home Menu.

Priority: High

When Available: When the player clicks to enter a queue.

Frequency of Use: Moderate, depending on matchmaking demand for users

Channel to Actor: UI notification

Secondary Actors: Matchmaking service

Channels to Secondary Actors: System-to-system communication with matchmaking service

Open Issues:

- Should players be matched based on certain skill level or expertise?
- Extension to above: If there isn’t a player that can match skill levels, can the user be notified if & when a match is found?

Use Case: Tic Tac Toe Game Instance

Iteration: 1

Primary Actor: Player

Goal In Context: Enable the player to play a Tic Tac Toe game against another player.

Preconditions: A match has been successfully set up and an opponent is found from queue.

Trigger: The system begins the game instance when a suitable opponent is

Scenario:

1. System pairs player with it's opponent and initiates the game instance.
2. Player then receives a notification that the game is starting and is presented with the Tic Tac Toe game board
3. Then, the player and their opponent take turns making moves until the game ends (either a win, loss or draw)
4. The system records outcome of the game and updates the leaderboard with the respective winner.
5. The player is then directed to the result screen which displays "victory", "defeat" or "draw".
6. Player can choose to return to the Tic Tac Toe home screen or view the leaderboard from the results page.

Post Conditions: Game results are then recorded, and the player is directed to the game aftermath options.

Exceptions:

- **E1:** If the opponent disconnects or withdraws from game, player is notified and the game ends
- **E2:** If there is a connection issue on the player's end, system will save & freeze the game

Priority: High

When Available: Once matchmaking is done.

Frequency of Use: Varies, based on player's success in finding an opponent.

Channel to Actor: UI

Secondary Actors: Opponent user

Channels to Secondary Actors: Network communication for Tic Tac Toe moves

Open Issues:

- Should players be able to send challenges or game requests to recent opponents or opponents who are higher up on leaderboard?

Use Case: No Match Found

Iteration: 1

Primary Actor: Player

Goal In Context: Tells the player that the system was unable to find a match for them & it then provides options for them to exit the game or retry.

Preconditions: The player has joined the Tic Tac Toe Game Queue.

Trigger: If the system is unable to find an opponent for the player

Scenario:

1. The player joins the Tic Tac Toe Game Queue.
2. System searches for an opponent
3. After some time (5 minutes for ex), the system fails to find a match.
4. The system displays a “No Match Found” pop-up to the player
5. User is then given options to try joining the queue again or return to the Tic Tac Toe Home Menu.

Post Conditions: No match was found and player can choose to exit, retry later or try again now

Exceptions:

- **E1:** If there is an error getting the player’s queue number, error message is shown & player is then redirected to Tic Tac Toe home menu

Priority: Medium

When Available: If the matchmaking queue times out or if there aren’t enough players in the queue

Frequency of Use: Occasional

Channel to Actor: UI notification

Secondary Actors: None

Channels to Secondary Actors: N/A

Open Issues:

- Would implementing a “waiting room” feature with instructions for the game or mini challenges be a good way to increase user retention?

Use Case: Tic Tac Toe Leaderboard

Iteration: 1

Primary Actor: Player

Goal In Context: enables the player to view leaderboard for Tic Tac Toe & shows rankings in order of top player scores

Preconditions:

- The player passed the Tic Tac Toe Home Menu.

Trigger: The player selects the “Leaderboard” option from the Tic Tac Toe Home Menu.

Scenario:

1. The player navigates to the Tic Tac Toe Home Menu.
2. The player selects “Leaderboard.”
3. The system retrieves the latest leaderboard data for Tic Tac Toe.
4. The system displays the leaderboard, showing the following statistics for each player:
    - **Rank**: position based on win rate/score.
    - **Username**: display name of ranked players.
    - **Total Wins**
    - **Win Rate**: percentage of games won over total games.
    - **Longest Win Streak**: longest number of wins in a row.
    - **Games Played**: total number of games the player has done.
    - **Last Played Date**: date of most recent game.
    - **Moves to Win (Average)**: average number of moves player needs to take to win.
5. The player can scroll through or search the leaderboard to find specific players or rankings.
6. The player can filter the leaderboard (longest streak, most games played etc.)
7. The player can return to the Tic Tac Toe Home Menu

Post Conditions: The player has successfully viewed the Tic Tac Toe leaderboard and can navigate back to other Tic Tac Toe options.

Exceptions:

- **E1:** if the leaderboard data fails to load due to a network issue, the system will display an error message and provide the option to retry or return to the Tic Tac Toe Home Menu.
- **E2:** if the leaderboard is down, then the system informs the player and directs back to the Tic Tac Toe Home Menu.

Priority: Medium

When Available: Anytime the player is at the Tic Tac Toe Home Menu.

Frequency of Use: Occasional

Channel to Actor: UI display

Secondary Actors: Leaderboard database/service

Channels to Secondary Actors: System-to-database communication to retrieve leaderboard information

Open Issues:

- Would adding achievements, badges or premium features increase player retention and engagement?

Use Case: Checkers Home Menu

Iteration: 1

Primary Actor: Player

Goal In Context: Player is at the Home Menu screen for Checkers, where they can choose to enter the queue to play the game or view the leaderboard.

Preconditions:

- The player has logged in and is at the Player Profile Home Menu.
- The Checkers game is available.

Trigger: The player chooses to play Checkers and clicks on Checkers.

Scenario:

1. The player has logged into their account.
2. They are now at the Player Profile Home Menu.
3. They choose to play Checkers and click onto the game.
4. They are brought to the Home Menu screen for Checkers.
5. They can now choose to view the leaderboard or join a queue.

Post Conditions: The player is directed to view the leaderboard for Checkers, or to join the queue for the game.

Exceptions:

- **E1:** The Checkers module is unavailable, system shows an error message and the player is redirected to the Player Profile Home Menu.

Priority: High

When Available: After the player logs in and arrives at the Checker Home Menu.

Frequency of Use: Frequent, shown every time the player wants to play Checkers.

Channel to Actor: UI

Secondary actors: None

Channels to secondary actors: N/A

Open Issues:

- Should there be more options for the game at the Home Menu for the player to choose from?

Use Case: Checkers game queue

Iteration: 1

Primary Actor: Player

Goal In Context: The player would be able to join a queue to get matched up with an opponent.

Preconditions:

- The player has logged in and is at the Checkers Home Menu.
- The matchmaking service is available.

Trigger: The player chooses “Join Queue” from the Checkers Home Menu.

Scenario:

1. The player has logged into their account.
2. They choose to play Checkers and selects “Join Queue”.
3. The matchmaking system searches for an opponent.
4. If they are matched up with another player, they are directed to the Checkers game instance.

Post Conditions: The player is successfully matched up with another player, or they could not be matched up because no match is found.

Exceptions:

- **E1:** No other player wants to play Checkers at the same time and matching up fails.
- **E2:** The matchmaking system is down, the system shows an error message and redirects the player to the Checkers Home Menu.

Priority: High

When Available: When a player chooses to join a queue.

Frequency of Use: Moderate, depends on the player’s choices and wait times.

Channel to Actor: UI

Secondary actors: Matchmaking service

Channels to secondary actors: System-to-system communication

Open Issues:

- Should the player be able to choose to cancel queuing and get back to the game Home Menu?

Use Case: Game instance

Iteration: 1

Primary Actor: Player

Goal In Context: The player can play the Checkers game with an opponent.

Preconditions: The player has logged in and is matched up successfully with another player.

Trigger: The player chooses to play Checkers and another player is found.

Scenario:

1. The player clicks “Join Queue” and is matched up successfully with another player.
2. They are now at a screen where they can play Checkers and are notified that the game has started.
3. The players take turns and play the game until the game ends (win, lose, or draw).
4. After the game, they will be directed to the aftermath screen.
5. They can choose to return to the Checkers Home Menu or view the leaderboard.

Post Conditions: The game results are recorded and the aftermath screen is shown after the game is finished.

Exceptions:

- **E1:** The opponent disconnects, the system notifies the player, the game is ended.
- **E2:** The player’s connection is lost, the system stops the game.

Priority: High

When Available: After matchmaking.

Frequency of Use: Depends if the player can successfully match up with an opponent.

Channel to Actor: UI with real-time updates

Secondary actors: Opponent player

Channels to secondary actors: Network communication

Open Issues:

- Should the player be able to rematch with the opponent?

Use Case: No match found

Iteration: 1

Primary Actor: Player

Goal In Context: The player is notified that there is no match currently and they can retry or exit.

Preconditions: The player chooses to join a Checkers Game Queue.

Trigger: The player waits in the queue but the system could not find an opponent with in a certain time period.

Scenario:

1. The player chooses to join a Checkers Game Queue.
2. The matchmaking service searches for an opponent.
3. After a timeout period, the system still could not find an opponent.
4. The player is shown the “No Match Found” notification.
5. They have options to retry or exit back to the Checkers Home Menu.

Post Conditions: The player can try again or they are directed back to the Home Menu.

Exceptions:

- **E1:** There is an error while getting the player’s queue status, the system shows an error message and the player is redirected back to the Checkers Home Menu.

Priority: Medium

When Available: When there are low numbers of players.

Frequency of Use: Occasional, depends on the number of players.

Channel to Actor: UI notification

Secondary actors: None

Channels to secondary actors: N/A

Open Issues:

- What if there is no match for the player for a very long time and the player really wants to play?
- Should we include a function to have a game with AI?

Use Case: Checkers leaderboard

Iteration: 1

Primary Actor: Player

Goal In Context: The player would be able to see a screen with the leaderboard data for Checkers.

Preconditions:

- The player chooses Checkers from their profile Home Menu.
- The leaderboard functionality is available.

Trigger: The player is at the Checkers Home Menu and chooses “Leaderboard”.

Scenario:

1. The player chooses Checkers on their Profile Home Menu.
2. They want to see the leaderboard and select “Leaderboard”.
3. They are shown with the leaderboard for Checkers with the following data:
    - **Rank:** The player’s ranking based on win rate or score.
    - **Username:** The display name of the player.
    - **Total Wins:** The total number of games the player has won.
    - **Win Rate:** The percentage of games won out of total games played.
    - **Longest Win Streak:** The highest number of consecutive wins of the player.
    - **Games Played:** The total number of games the player has played.
    - **Last Played Date:** The date when the player last played.
    - **Moves to Win (Average):** The average number of moves the player used to win a game.

Post Conditions: The player has now viewed the leaderboard for Checkers with their stats and also other players’ stats, they can now go back to the Checkers Home Menu.

Exceptions:

- **E1:** The system fails to retrieve leaderboard data, the player is shown an error message.
- **E2:** The leaderboard functionality is not available, the player is informed and the have the choice to come back later or return to the Checkers Home Menu.

Priority: Medium

When Available: When the player is at the Checkers Home Menu.

Frequency of Use: Occasional, depends on if the player wants to check the leaderboard.

Channel to Actor: UI display of the leaderboard

Secondary actors: Leaderboard database/service

Channels to secondary actors: System-to-database communication

Open Issues:

- Should there be a friends system?
- Should the player be able to sort or filter the leaderboard?